

A Two-by-Two Counterexample to Zou’s Heinz–Heron Norm Conjecture

Math discovery packet

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Outcome. The conjecture in Remark 4.2 of L. Zou, *J. Math. Inequal.* 7 (2013), 389–397, is false. A real 2×2 counterexample uses the trace norm, $\nu = 17/20$, and data whose entries and logarithmic spectra are all rational tenths. An exact-rational interval certificate proves a strict gap greater than $1001/10^6$.

1 The conjecture and a transcription issue

For positive matrices A, B , an arbitrary matrix X , and $0 \leq \nu \leq 1$, set

$$\alpha(\nu) = 1 - 4(\nu - \nu^2) = (2\nu - 1)^2.$$

Zou conjectured that every unitarily invariant norm satisfies

$$\begin{aligned} \frac{1}{2} \|A^\nu X B^{1-\nu} + A^{1-\nu} X B^\nu\| &\leq (1 - \alpha(\nu)) \|A^{1/2} X B^{1/2}\| \\ &\quad + \alpha(\nu) \left\| \frac{AX + XB}{2} \right\|. \end{aligned} \tag{1}$$

This is inequality (4.2) and the Conjecture in Remark 4.2 of the included 2013 primary paper.

The lane target is Y. Kapil, C. Conde, M. S. Moslehian, M. Singh, and M. Sababheh, *Norm inequalities related to the Heron and Heinz means*, arXiv:1709.09813. In its “On Zou’s questions” paragraph it prints the second Heinz term as $B^\nu X A^{1-\nu}$. Comparison with Zou’s original display shows that this is a transcription error. The crossed version is already false at $\nu = 1/2$ by elementary matrix-unit examples. The result below addresses the mathematically intended and correctly sourced conjecture (1), not that typo.

2 Explicit counterexample

Let $\|M\|_1 = \text{tr}((M^*M)^{1/2})$, a unitarily invariant norm. Put

$$\nu = \frac{17}{20}, \quad s = 2\nu - 1 = \frac{7}{10}, \quad \alpha(\nu) = s^2 = \frac{49}{100},$$

and define

$$A = \begin{pmatrix} e^{3/5} & 0 \\ 0 & e^{13/5} \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & e^{9/5} \end{pmatrix}, \quad T = \frac{1}{10} \begin{pmatrix} 5 & -8 \\ -1 & 4 \end{pmatrix}.$$

Finally, let

$$X = A^{-1/2} T B^{-1/2}.$$

The matrices A, B are positive definite and X is a real 2×2 matrix.

Theorem 1. *For these data, inequality (1) fails for the trace norm. In fact, its left-hand side exceeds its right-hand side by more than $1001/10^6$.*

Proof. Write

$$h = \left(\frac{3}{10}, \frac{13}{10} \right), \quad k = \left(0, \frac{9}{10} \right), \quad d_{ij} = h_i - k_j.$$

Since $A_{ii} = e^{2h_i}$ and $B_{jj} = e^{2k_j}$, direct entrywise calculation gives

$$\begin{aligned} Y &:= \frac{1}{2} (A^\nu X B^{1-\nu} + A^{1-\nu} X B^\nu), & Y_{ij} &= \cosh(sd_{ij})T_{ij}, \\ A^{1/2} X B^{1/2} &= T, \\ Z &:= \frac{AX + XB}{2}, & Z_{ij} &= \cosh(d_{ij})T_{ij}. \end{aligned}$$

Thus, explicitly,

$$Y = \begin{pmatrix} \frac{1}{2} \cosh(21/100) & -\frac{4}{5} \cosh(21/50) \\ -\frac{1}{10} \cosh(91/100) & \frac{2}{5} \cosh(7/25) \end{pmatrix},$$

and

$$Z = \begin{pmatrix} \frac{1}{2} \cosh(3/10) & -\frac{4}{5} \cosh(3/5) \\ -\frac{1}{10} \cosh(13/10) & \frac{2}{5} \cosh(2/5) \end{pmatrix}.$$

For a real 2×2 matrix M , its singular values give the identity

$$\|M\|_1^2 = \|M\|_F^2 + 2|\det M|. \quad (2)$$

The exact interval calculation described below proves

$$\|Y\|_1 > 1.178, \quad \|T\|_1 < 1.1402, \quad \|Z\|_1 < 1.2153. \quad (3)$$

It follows that the right-hand side of (1) is strictly less than

$$\frac{51}{100}(1.1402) + \frac{49}{100}(1.2153) = 1.176999,$$

whereas its left-hand side is greater than 1.178. Their difference is therefore greater than $1.178 - 1.176999 = 0.001001 = 1001/10^6$. $\square$

3 Exact certificate for the three norm bounds

For completeness, here is a reproducible proof of (3). If $x \in \mathbb{Q}$ and $N = 24$, define

$$L_N(x) = \sum_{j=0}^N \frac{x^{2j}}{(2j)!}, \quad t_{N+1}(x) = \frac{x^{2N+2}}{(2N+2)!},$$

and

$$q_N(x) = \frac{x^2}{(2N+3)(2N+4)}.$$

The positive Taylor series and the decreasing ratio of its omitted terms give the rational enclosure

$$L_N(x) \leq \cosh x \leq L_N(x) + \frac{t_{N+1}(x)}{1 - q_N(x)}. \quad (4)$$

All arguments appearing in Y and Z are rational, so interval arithmetic using (4), followed by (2), is exact over $\mathbb{Q}$. It yields the following strictly positive margins:

Certified comparison	Rational-interval margin
$\ Y\ _1^2 > (1.178)^2$	greater than 2.72327×10^{-4}
$(1.1402)^2 > \ T\ _1^2$	exactly 5.604×10^{-5}
$(1.2153)^2 > \ Z\ _1^2$	greater than 3.18739×10^{-4}

The determinant intervals are positive as well: $\det Y > 0.08668$ and $\det Z > 0.03910$, justifying removal of the absolute values in (2). The included `verify_counterexample.py` performs every enclosure and comparison with Python’s exact `Fraction` arithmetic; floating point is used only to print readable decimal summaries.

For orientation, ordinary numerical evaluation gives

$$\|Y\|_1 \approx 1.1781155833, \quad \|T\|_1 \approx 1.1401754251, \quad \|Z\|_1 \approx 1.2151688567,$$

and an actual gap of approximately 0.0011933767.

4 Why the counterexample works

After conjugating away the geometric term, the conjecture asks a quadratic interpolation inequality for the Schur multipliers $T_{ij} \mapsto \cosh(s(h_i - k_j))T_{ij}$. Scalar Heinz–Heron interpolation controls each entry separately, but a unitarily invariant norm also couples the entries through singular values. In dimension two, formula (2) makes the extra coupling explicit: the determinant contribution bends differently at the intermediate parameter $s = 0.7$ than at $s = 0$ and $s = 1$. Independent left and right spectra create four distinct logarithmic gaps, and the intermediate trace norm lies above the conjectured quadratic chord.

This also explains why the Hilbert–Schmidt evidence motivating the conjecture does not settle it: the Frobenius part is entrywise quadratic, whereas the trace norm contains the additional determinant term.

5 Provenance, novelty, and scope

The exact 2013 source PDF, the 2017 target PDF, and a related 2018 paper are included. Searches by arXiv id, title, exact conjecture wording, formula fragments, author/title citations, and the hyperbolic-kernel formulation found no prior counterexample or explicit resolution of Zou’s Conjecture (4.2). This is a bounded novelty check, not a claim that the entire literature has been exhaustively certified.

The result has deliberately narrow scope:

- it refutes the universal unitarily invariant norm conjecture using the trace norm already in dimension two;
- it does not contradict Zou’s Hilbert–Schmidt theorem;
- it does not concern the separate positive-definite-function question, which arXiv:1709.09813 answers negatively;
- it corrects, but does not rely on, the crossed-exponent transcription in arXiv:1709.09813.

6 References

1. L. Zou, *Inequalities related to Heinz and Heron means*, J. Math. Inequal. 7 (2013), 389–397, doi:10.7153/jmi-07-34.
2. Y. Kapil, C. Conde, M. S. Moslehian, M. Singh, and M. Sababheh, *Norm inequalities related to the Heron and Heinz means*, Mediterr. J. Math. 14 (2017), Article 213; arXiv:1709.09813.
3. F. Gao and X. Ma, *Norm inequalities related to the Heinz means*, J. Inequal. Appl. 2018 (2018), Article 303.